Order Has Consequences: Total Effects and Fidelity-Conditioned Prediction Instability under Compound Image Corruptions

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Repository: https://github.com/aviral199/robustlens • **Data Artifacts:** 973,400 forward passes on 3,925 Imagenette images

Abstract— Deep neural networks for vision are widely evaluated under synthetic distribution shifts, but standard benchmarks treat corruptions as isolated unary operations (ImageNet-C) or along fixed unidirectional pipelines (R-Bench). In real-world imaging pipelines, degradations occur sequentially through multi-stage physical and digital processes governed by non-commutative operators ($T_A \blacksquare T_B \neq T_B \blacksquare T_A$). We conduct an extensive paired evaluation of the full symmetric group of three canonical degradations—Gaussian Noise (*N*), Defocus Blur (*B*), and JPEG Compression (*J*)—yielding all 3! = 6 permutation sequences across 5 severity levels evaluated on four prominent vision architectures (ResNet-50, ConvNeXt-Tiny, ViT-Base/16, Swin-Tiny), comprising 973,400 total forward passes across two distinct experimental regimes (486,700 records per regime on 3,925 FastAI Imagenette-v2 images with seed-matched noise realizations). We formulate two distinct estimands: (1) **Total Operational Effect** at standard ImageNet-C parameters, where operator non-commutativity drives cross-sequence top-1 accuracy gaps of up to 21.55% (ResNet-50) and 20.45% (ConvNeXt-Tiny) via downstream terminal smoothing; and (2) **Fidelity-Conditioned Comparison**, where pre-calibration targeting common (PSNR, SSIM) support contracts the cross-sequence range of the overall model mean by 63.18% (dropping from 18.43% down to 6.79%, while the average of the four individual architecture gap contractions is 58.08%, contracting by 77.15% in ConvNeXt and 72.62% in ResNet). Crucially, prespecified tolerances (±0.75 dB PSNR, ±0.035 SSIM) were not achieved (PSNR RMSE = 0.943 dB, SSIM RMSE = 0.1380), establishing this regime strictly as an approximate-fidelity descriptive comparison. Beyond aggregate accuracy, we formulate a complete 5-state joint prediction space partition (E_{11}, E_{10}, E_{01}, E_{00}^{same}, E_{00}^{diff}) and prove that **Predicted-Label Churn** ($P(Y\blacksquare_A \neq Y\blacksquare_B)$) is more than double **Correctness-State Churn** ($P(C_A \neq C_B) = L + G$), because Error Migration (E_{00}^{diff}) averages 23.85% in nominal pipelines and 22.40% under approximate fidelity. Pairs with near-identical accuracy ($|\Delta\text{Acc}| < 2.5\%$) exhibit up to 10.97% excess churn and over 40% label churn, proving that sequence order actively reshuffles classifier decision boundaries across identical visual instances.

1. Introduction & The Two Distinct Estimands

Benchmarking neural network robustness to distribution shift is essential for safety-critical deployment. While ImageNet-C (Hendrycks & Dietterich, 2019) established standard protocols for evaluating 15 isolated degradations, physical imaging pipelines (e.g., in autonomous driving or medical imaging) subject images to multi-stage physical and digital degradations. Because the mathematical operators governing image formation (spatial convolution, stochastic noise addition, and block DCT quantization) are **non-commutative** ($T_A \blacksquare T_B \neq T_B \blacksquare T_A$), the sequence order in which operators act creates distinct physical signals and downstream feature representations (Liao et al., 2020).

When investigating sequence order, conflating experimental goals leads to profound misinterpretation. We explicitly formalize two distinct estimands:

• **Estimand 1 (Total Operational Effect at Fixed Parameters):** Evaluates the net impact of sequence order when operators run at identical standard nominal settings. Here, low-level fidelity disparities (e.g., higher PSNR when blur follows JPEG) are the direct physical consequence of operator non-commutativity in real pipelines.

• **Estimand 2 (Fidelity-Conditioned Descriptive Comparison):** Evaluates whether cross-sequence accuracy disparities persist when operator parameters are calibrated to target common (PSNR, SSIM) support. Because discrete calibration did not satisfy prespecified tolerances (±0.75 dB PSNR, ±0.035 SSIM), this estimand provides a *descriptive comparison under approximate common support*, rather than an exact causal effect at matched fidelity. Confounding was reduced, but not eliminated.

2. Related Work & Novelty Demarcation

Prior benchmarks predominantly evaluate unary corruptions (ImageNet-C) or temporal video sequences (ImageNet-P). Dodge & Karam (2016, 2019) pioneered psychometric distortion matching on single corruptions. Goyal et al. (2022) studied dual corruptions (noise and motion blur in low-light imaging), noting order asymmetry; however, they evaluated only 2! = 2 orderings without calibrating parameters to common support across the full permutation group. Li et al. (2025) introduced R-Bench, evaluating a 7-stage physical pipeline on MLLMs, but followed a single unidirectional acquisition sequence without permuting stages. Liao et al. (2020) demonstrated in digital forensics that 2-operator chains leave detectable forensic traces, confirming operator non-commutativity. Wei et al. (2026) showed that paired single-image evaluations uncover classwise correctness bounds under unary visual perturbations. Prediction churn was formalized across model retraining iterations by Milani Fard, Cormier, Canini, & Gupta (2016). RobustLens provides the first full-factorial 3! = 6 permutation study on identical physical images with seed-matched noise, formulating a 5-state prediction space partition that decouples predicted-label churn from correctness churn, a full-rank 6-cell decomposition, and an empirical fidelity-calibration audit documenting gap contraction.

3. Mathematical Framework & Full 5-State Prediction Partition

Let x be a clean RGB image. Operators are Gaussian Noise (N_σ), Defocus Blur ($B_{\kappa,\gamma}$), and JPEG Compression (J_q). The symmetric group S_3 contains 6 sequences: NBJ, NJB, BNJ, BJN, JNB, JBN (applied left to right; e.g., NBJ ends in JPEG, NJB ends in Blur, BJN ends in Noise).

Let $\blacksquare_A$, $\blacksquare_B$ be predicted classes, Y ground truth, and C_A, $C_B \in \{0, 1\}$ correctness indicators. The joint prediction space partitions into five mutually exclusive, exhaustive transitions:

1. E_{11} (Both Correct): $\blacksquare_A = Y \wedge \blacksquare_B = Y$; 2. E_{10} (Loss / *L*): $\blacksquare_A = Y \wedge \blacksquare_B \neq Y$; 3. E_{01} (Gain / *G*): $\blacksquare_A \neq Y \wedge \blacksquare_B = Y$; 4. E_{00}^{same} (Same Wrong Class): $\blacksquare_A = \blacksquare_B \neq Y$; 5. E_{00}^{diff} (Error Migration / Different Wrong Classes): $\blacksquare_A \neq Y \wedge \blacksquare_B \neq Y \wedge \blacksquare_A \neq \blacksquare_B$.

Conservation of probability holds: $P(E_{11}) + P(E_{10}) + P(E_{01}) + P(E_{00}^{\text{same}}) + P(E_{00}^{\text{diff}}) = 1$.

We define and distinguish two churn metrics:

• **Correctness-State Churn:** $P(C_A \neq C_B) = P(E_{10}) + P(E_{01}) = L + G$

• **Predicted-Label Churn:** $P(\blacksquare_A \neq \blacksquare_B) = L + G + P(E_{00}^{\text{diff}})$ = Correctness Churn + Error Migration.

• **Agreement Rate:** $P(\blacksquare_A = \blacksquare_B) = P(E_{11}) + P(E_{00}^{\text{same}}) = 1$ - Label Churn.

• **Excess Churn:** Excess Correctness Churn = $2 \cdot \min(G, L) = (G+L) - |\Delta\text{Acc}|$; Excess Label Churn = $2 \cdot \min(G, L) + P(E_{00}^{\text{diff}})$.

Theorem: Whenever models make distinct incorrect predictions ($P(E_{00}^{\text{diff}}) > 0$), Correctness Churn strictly understates classifier instability. In our experiments, Error Migration is ~23%, causing Predicted-Label Churn to be more than double Correctness Churn.

4. Full-Rank Saturated Six-Cell Linear Model & Statistical Inference

To decompose sequence variance across all six cell means $\mu = [\mu_{NBJ}, \mu_{NJB}, \mu_{BNJ}, \mu_{BJN}, \mu_{JNB}, \mu_{JBN}]^T$, we specify the linear model $\mu = X\beta$ with reference levels Start Noise and End Noise:

Columns of X: [Intercept, Start Blur, Start JPEG, End Blur, End JPEG, Parity Contrast (Even: +1, Odd: -1)]. Because $det(X) = -6 \neq 0$, X has full rank 6, uniquely determining $\beta = X^{-1}\mu$.

Critical Design Properties:

1. **Non-Orthogonal Design:** $X^T X \neq I$; reference-coded parameters must be interpreted alongside direct marginal contrasts.
2. **Unobserved Reference Condition:** No 3-operator sequence without repetition can both start and end with Noise. Hence, Intercept β_0 is an extrapolated linear parameter, not an observable baseline.
3. **Balanced Parity Contrast:** Even permutations (A_3: NBJ, BJN, JNB) and Odd permutations (NJB, BNJ, JBN) both contain every operator exactly once as an entry operator and exactly once as an exit operator. The coefficient $\beta_5 = (\mu\blacksquare_{Even} - \mu\blacksquare_{Odd})/2$ is perfectly balanced against entry and exit marginals, with full contrast $2\beta_5$.
4. **Clustered Bootstrap & Multiplicity:** We perform clustered bootstrap resampling ($B = 5{,}000$ clusters on 3,925 physical image IDs). Asymptotic two-sided Wald p-values are computed as $log\ p = log(2) + logsf(|z|)$ using cluster-bootstrap standard errors. Holm step-down adjustment controls FWER ($\alpha = 0.05$) on primary contrasts; Benjamini-Hochberg controls FDR ($q = 0.05$) on secondary families.

5. Regime 1: Nominal-Order Benchmark (Estimand 1)

Under fixed nominal parameters (ImageNet-C severities 1–5), models evaluated on identical physical images with matched noise seeds exhibit massive performance gaps.

Model	Arch	NBJ	NJB	BNJ	BJN	JNB	JBN	Mean	Min	Max	Gap (Δ)	Brier
ResNet-50	CNN	53.44 %	56.88 %	37.30 %	35.71 %	57.27 %	35.79 %	46.06%	35.71 %	57.27 %	**21.55%**	0.1587
ConvNeXt-T	Modern CNN	63.98 %	66.44 %	48.63 %	45.99 %	65.44 %	46.33 %	56.14%	45.99 %	66.44 %	**20.45%**	0.1265
ViT-B/16	Transformer	58.39 %	60.14 %	39.77 %	41.64 %	59.19 %	42.23 %	50.23%	39.77 %	60.14 %	**20.37%**	0.1481
Swin-Tiny	Hier. ViT	59.81 %	61.78 %	45.81 %	50.24 %	62.94 %	50.63 %	55.20%	45.81 %	62.94 %	**17.13%**	0.1286
Overall Mean	—	58.90 %	61.31 %	42.88 %	43.40 %	61.21 %	43.75 %	51.91%	42.88 %	61.31 %	**18.43%**	0.1405

Table 1: Nominal-Order Top-1 Accuracy & Brier Scores. Evaluated across 3,925 images (19,625 evaluations per sequence per model; 471,000 compound records).

Direct Marginal Contrasts: Start Blur vs. Noise: -16.97% ($p < 10^{-100}$); Start JPEG vs. Noise: -7.63% ($p < 10^{-100}$). End Blur vs. Noise: **+17.69%** ($p < 10^{-100}$); End JPEG vs. Noise: +7.32% ($p < 10^{-100}$). Balanced Parity Contrast: **+5.19%** ($2\beta_5$, $p < 10^{-100}$).

Finding: In nominal evaluations, the massive 18.43% sequence gap is dominated by terminal operator identity: ending in Blur confers a +17.69% marginal boost over terminal noise because spatial convolution attenuates prior high-frequency noise and DCT block artifacts, boosting PSNR by ~6.2 dB (mean PSNR 24.6 dB for terminal blur vs. 18.3 dB for terminal noise).

6. Regime 2: Approximate-Fidelity Descriptive Comparison (Estimand 2)

To test whether order sensitivity persists when output distortion is constrained, operator parameters were pre-calibrated on an independent training partition ($N_{cal} = 50$) targeting five common (PSNR, SSIM) support bins across sequences.

Target	Target PSNR	Target SSIM	Max Abs PSNR Resid	PSNR Tol (≤ 0.75 dB)	Max Abs SSIM Resid	SSIM Tol (≤ 0.035)
Target 1	28.0 dB	0.850	1.117 dB (NJB)	**Failed**	0.0686 (BJN)	**Failed**
Target 2	24.5 dB	0.740	0.936 dB (BJN)	**Failed**	0.1758 (BJN)	**Failed**
Target 3	21.5 dB	0.620	1.260 dB (NBJ)	**Failed**	0.2379 (BJN)	**Failed**
Target 4	18.5 dB	0.480	1.989 dB (NBJ)	**Failed**	0.2373 (BJN)	**Failed**
Target 5	16.0 dB	0.360	2.639 dB (NBJ)	**Failed**	0.2067 (BJN)	**Failed**
Overall	—	—	**2.639 dB (Max)**	**RMSE: 0.943 dB (Failed)**	**0.2379 (Max)**	**RMSE: 0.1380 (Failed)**

Table 2: Measured Calibration Residuals and Tolerance Violations ($N_{cal} = 50$). On the 3,925 validation images, cross-sequence PSNR spread reached 3.10 dB at Target 5, and SSIM spread reached 0.359. Prespecified tolerances were not achieved over the tested discrete grid. This failure stems from the functional trade-off between smoothing and edge preservation in discrete parameter spaces. It does not prove universal physical impossibility across continuous spaces, but it confirms that this regime is strictly an **approximate-fidelity comparison**.

Model	NBJ	NJB	BNJ	BJN	JNB	JBN	Matched Mean	Matched Gap	Nominal Gap	Contraction	Mean Brier
ResNet-50	50.15 %	46.76 %	45.18 %	50.63 %	47.05 %	51.08 %	48.47%	**5.90%**	21.55%	**72.62%**	0.1558
ConvNeXt-T	60.65 %	56.46 %	55.97 %	59.22 %	56.47 %	59.86 %	58.11%	**4.67%**	20.45%	**77.15%**	0.1245
ViT-B/16	56.54 %	53.39 %	46.51 %	54.72 %	53.01 %	55.68 %	53.31%	**10.03%**	20.37%	**50.75%**	0.1444
Swin-Tiny	57.61 %	50.44 %	53.93 %	61.23 %	52.31 %	62.12 %	56.27%	**11.68%**	17.13%	**31.80%**	0.1433
Overall Mean	56.23 %	51.76 %	50.40 %	56.45 %	52.21 %	57.18 %	54.04%	**6.79%**	18.43%	**63.18%**	0.1420

Table 3: Top-1 Accuracy, Brier Scores, and Gap Contraction under Approximate Fidelity.
Direct Marginal Contrasts: Start Blur vs. Noise: -0.58% ($p = 0.089$); Start JPEG vs. Noise: +0.70% ($p = 0.041$). End Blur vs. Noise: **-4.83%** ($p < 10^{-100}$); End JPEG vs. Noise: -3.50% ($p < 10^{-100}$). Balanced Parity Contrast: **+1.85%** ($2\beta_5$, $p < 10^{-100}$).
Key Insights: (1) *Dramatic Gap Contraction:* Equalizing low-level distortion contracts the cross-sequence range of the overall model mean from 18.43% down to 6.79% (a 63.18% reduction in the overall mean range; the arithmetic average of the four individual architecture gap contractions is 58.08%, contracting by 77.15% in ConvNeXt, 72.62% in ResNet, 50.75% in ViT, and 31.80% in Swin). (2) *Reversal of the End Operator Effect:* End Blur flips from +17.69% to **-4.83%**. When low-level PSNR is balanced, terminal blur attenuates discriminative edges without conferring an artifact-smoothing advantage. (3) *Architectural Divergence:* ConvNeXt and ResNet contract by >70%, whereas Swin-Tiny retains an 11.68% residual gap (contracting by only 31.80%), indicating elevated sensitivity to multi-stage spatial operations.

7. Saturated Linear Decomposition Parameters

Table 4 compares the full-rank saturated six-cell decomposition parameters across both experimental regimes ($B = 5{,}000$ clustered bootstrap resamples, Holm FWER-adjusted).

Parameter	Nominal Estimate (%)	Nominal 95% Clustered CI	Nominal p_{Holm}	Matched Estimate (%)	Matched 95% Clustered CI	Matched p_{Holm}
β_0 **(Intercept)**	+51.63	[+50.71, +52.52]	$< 10^{-100}$	+59.51	[+58.51, +60.51]	$< 10^{-100}$
β_1 **(Start Blur)**	-10.83	[-11.14, -10.53]	$< 10^{-100}$	-3.99	[-4.22, -3.75]	$< 10^{-100}$
β_2 **(Start JPEG)**	-5.29	[-5.50, -5.08]	$< 10^{-100}$	-1.40	[-1.58, -1.23]	$< 10^{-50}$
β_3 **(End Blur)**	**+12.27**	[+11.88, +12.67]	$< 10^{-100}$	**-6.82**	[-7.18, -6.46]	$< 10^{-100}$
β_4 **(End JPEG)**	**+4.67**	[+4.35, +5.00]	$< 10^{-100}$	**-4.20**	[-4.50, -3.91]	$< 10^{-100}$
β_5 **(Parity Contrast)**	**+2.60**	[+2.52, +2.67]	$< 10^{-100}$	**+0.93**	[+0.86, +0.99]	$< 10^{-100}$

Table 4: Full-Rank Saturated Six-Cell Decomposition Coefficients. Reference levels: Start Noise, End Noise. Point estimates and CIs in % Top-1 Accuracy. Even after conditioning on entry and exit operators, the balanced parity contrast β_5 remains statistically significant in both regimes (+2.60% nominal, +0.93% matched, $p < 10^{-100}$).

8. Prediction Instability, Error Migration & Per-Class Breakdown

A cornerstone empirical finding is that **aggregate accuracy obscures massive per-image prediction churn and error migration.**

Regime	Model	Both Corr (E_{11})	Loss (E_{10})	Gain (E_{01})	Same Wrong (E_{00}^{same})	Error Mig (E_{00}^{diff})	Corr Churn	Excess Churn	Label Churn	Agree Rate	Mean |ΔConf|
Nominal	ConvNeXt	47.63%	11.64%	5.38%	14.99%	**20.36%**	17.02%	5.36%	**37.38%**	62.62%	0.1662
Nominal	ResNet-50	36.96%	12.06%	6.15%	14.90%	**29.93%**	18.20%	5.72%	**48.14%**	51.86%	0.1850
Nominal	Swin-Tiny	47.98%	8.49%	5.96%	16.06%	**21.51%**	14.45%	5.82%	**35.96%**	64.04%	0.1835
Nominal	ViT-Base	41.20%	11.76%	6.31%	17.17%	**23.57%**	18.06%	6.68%	**41.63%**	58.37%	0.1732
Nominal	**Overall**	**43.44%**	**10.99%**	**5.95%**	**15.78%**	**23.85%**	**16.93%**	**5.90%**	**40.78%**	**59.22%**	**0.1770**
Matched	ConvNeXt	52.64%	5.49%	5.44%	17.44%	**19.00%**	10.93%	8.51%	**29.92%**	70.08%	0.1163
Matched	ResNet-50	41.92%	6.18%	6.92%	15.54%	**29.44%**	13.10%	10.15%	**42.54%**	57.46%	0.1481
Matched	Swin-Tiny	50.31%	4.78%	7.14%	18.03%	**19.74%**	11.92%	5.99%	**31.66%**	68.34%	0.1307
Matched	ViT-Base	46.61%	6.61%	6.79%	18.55%	**21.44%**	13.40%	9.44%	**34.85%**	65.15%	0.1314
Matched	**Overall**	**47.87%**	**5.76%**	**6.57%**	**17.39%**	**22.40%**	**12.34%**	**8.52%**	**34.74%**	**65.26%**	**0.1316**

Table 5: Complete 5-State Joint Prediction Space Transitions. Averaged across all 15 pairwise order comparisons per model.
Finding 1: Label Churn is More Than Double Correctness Churn: Across all architectures, Error Migration (E_{00}^{diff}) averages 23.85% (nominal) and 22.40% (matched). Because models frequently transition between different incorrect classes across sequence permutations, inspecting only correctness churn ignores over 60% of total prediction turnover.
Finding 2: Decoupling of Aggregate Accuracy and Error Churn: Condition pairs with near-zero accuracy difference (e.g., ConvNeXt matched NJB vs. JNB, |ΔAcc| = 0.01%) exhibit 8.23% flipped predictions (4.12% gain, 4.11% loss), yielding 8.22% excess churn across 1,613 physical images.

Class Name	Nom Acc	Nom Conf	Nom Brier	Nom Label Churn	Nom Corr Churn	Nom Error Mig	Matched Acc	Matched Label Churn	Matched Error Mig
tench	73.57%	66.25%	0.0921	29.09%	17.56%	11.54%	77.22%	21.54%	11.91%
golf ball	72.45%	66.65%	0.1088	28.86%	17.89%	10.97%	75.07%	21.88%	10.83%
parachute	68.01%	61.06%	0.1224	31.19%	16.22%	14.97%	68.97%	28.00%	15.75%
chain saw	55.56%	51.44%	0.1170	41.93%	17.42%	24.51%	59.32%	34.59%	22.64%
French horn	51.98%	52.58%	0.1053	41.77%	14.21%	27.56%	53.81%	36.00%	27.08%
garbage truck	49.61%	50.66%	0.1528	43.23%	19.56%	23.67%	51.25%	36.10%	21.24%
gas pump	48.57%	48.14%	0.1253	46.52%	17.51%	29.01%	50.56%	40.01%	28.27%
cassette player	41.35%	42.38%	0.1988	45.62%	20.64%	24.98%	44.40%	39.95%	22.40%
English springer	40.48%	40.22%	0.1786	48.72%	17.85%	30.87%	41.61%	42.85%	28.25%
church	**18.35%**	38.21%	0.2065	**50.45%**	**11.04%**	**39.41%**	19.22%	**46.07%**	**34.70%**

Table 6: Per-Class Vulnerability Breakdown Across 10 Imagenette Classes. For 'church', accuracy collapses to 18.35% and correctness churn is only 11.04%. If an evaluation only inspected correctness churn, it would classify 'church' as the *most stable* class! However, Error Migration is an astounding **39.41%**, pushing Predicted-Label Churn to **50.45%**. Under sequence permutations, the classifier constantly rotates between incorrect classes.

9. Calibration & Expected Calibration Error (ECE)

Expected Calibration Error (ECE) was computed using 10 equal-width bins directly from raw softmax confidence scores and correctness indicators: $ECE = \sum_{m=1}^{M} (|B_m| / N) |acc(B_m) - conf(B_m)|$.

• **Overconfidence under Severe Shift:** Mean confidence degrades substantially slower than accuracy. Under Target 5, ResNet-50 achieves 31.2% accuracy while maintaining 54.8% confidence (ECE > 22%).

• **Terminal Noise Maximizes Miscalibration:** Sequences ending in Gaussian Noise (BJN, JBN) produce the highest calibration error (up to 18.58% in Swin-Tiny). High-frequency noise perturbs localized activation maps while preserving softmax peakiness.

10. Scope Boundaries, Limitations & Open Science

In accordance with rigorous academic integrity standards, we explicitly delineate the boundaries of this benchmark:

1. **Approximate Fidelity, Not Exact Matching:** Prespecified calibration tolerances (±0.75 dB PSNR, ±0.035 SSIM) were not achieved over the tested discrete grid. Residual PSNR spread reached 3.10 dB at Target 5. Residual performance differences cannot be interpreted as pure architectural effects without acknowledging residual distortion imbalance.
2. **Finite Grid Boundary:** The failure of our discrete parameter search does not prove universal physical impossibility across continuous spaces or alternate corruption formulations.
3. **Dataset & Task Scope:** Evaluated on FastAI Imagenette-v2 (10 classes, 3,925 images). Evaluations on full ImageNet-1k and dense prediction tasks (detection, segmentation) remain necessary before generalizing benchmark conclusions.
4. **Code & Data Availability:** All data artifacts (486,700 records per regime, totaling 973,400 forward passes) and reproducible scripts are maintained in the repository at https://github.com/aviral199/robustlens.

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